Class Test 3 **Date: November 25, 2024**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-2/T-2 CSE 219: Signals and Linear Systems

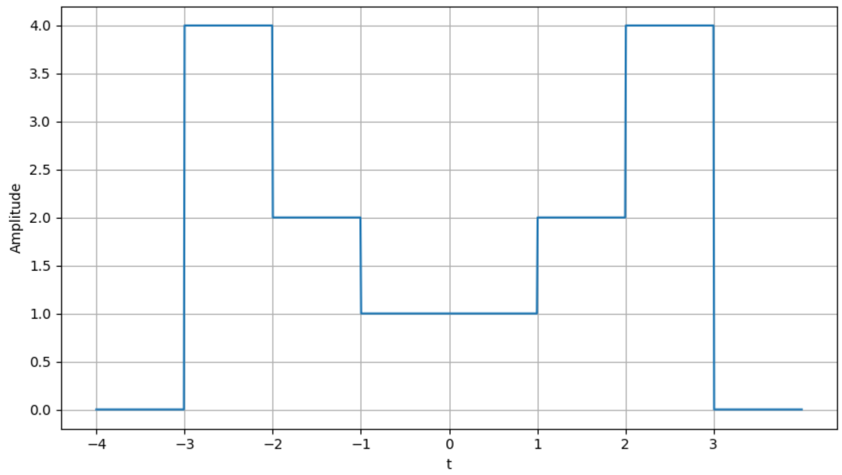
Time: 25 minutes **Marks: 20**

Student Name: ______________________________ Student No: ____________

1. Find fourier transform of the signal $x(t) = te^{2t}u(3-t)$

(8 Marks)

2. Use linearity property to find fourier transform of the signal given in the plot below. You are not allowed to use the fourier transform formula directly. (7 Marks)



3. Given $X_1(f) = sinc(f)$, and $X_2(f) = sinc^2(f)$ Find the inverse fourier transform of $X_1(f) * X_2(f)$ (where * denotes convolution operator) (5 Marks)